LEMBAR ANALISA

Praktikum Jaringan Komputer -1 (Analisa Paket Layer)
Tanggal Praktikum : Senin, 2 Maret 2026
Kelas : 2 TRI B
Nama dan NRP : - Raden Muhammad Davino Arya Cahyakesuma (2425600045)
- Muhammad Slamet Ridho Novianto (2425600050)

A. Catat no ip komputer sumber dan target, dan catat juga hasil perintah : arp -a (poin 1)

```
arz@RaidenArya:~$ arp -a
_gateway (10.85.153.85) at e2:8d:90:ce:4b:61 [ether] on wlp3s0
RaidenArya (10.85.153.130) at <incomplete> on wlp3s0
```

```
root@vian-ubuntu:~# arp -a
? (10.85.153.130) at 2c:9c:58:24:1c:21 [ether] on wlp3s0
? (10.85.153.129) at <incomplete> on wlp3s0
_gateway (10.85.153.85) at e2:8d:90:ce:4b:61 [ether] on wlp3s0
```

IP Target: 10.85.153.96
IP Sumber: 10.85.153.130
Analisa

- Arp -a menampilkan ARP cache yang sudah pernah dikenali OS. ARP cache bertambah kalau ARP request dapat jawaban dan diatur dinamis oleh kernel.
- Kalau sebelum ping belum komunikasi dengan IP target, entri target sering belum muncul.

B. Catat hasil perintah arp –a setelah dilakukan ping, catat juga proses yang terjadi di Wireshark (poin 4)

```
arz@RaidenArya:~$ ping 10.85.135.96
PING 10.85.135.96 (10.85.135.96) 56(84) bytes of data.
^C
--- 10.85.135.96 ping statistics ---
13 packets transmitted, 0 received, 100% packet loss, time 12263ms

arz@RaidenArya:~$
```

No.	Time	Source	Destination	Protocol	Length	Info
215	59.236558966	146.190.225.48	10.85.153.129	TCP	54	80 → 33360
216	59.340780960	10.85.153.129	10.85.135.96	ICMP	98	Echo (ping)
217	60.364972256	10.85.153.129	10.85.135.96	ICMP	98	Echo (ping)
218	61.388678812	10.85.153.129	10.85.135.96	ICMP	98	Echo (ping)
219	62.413189683	10.85.153.129	10.85.135.96	ICMP	98	Echo (ping)
220	63.436773575	10.85.153.129	10.85.135.96	ICMP	98	Echo (ping)
221	64.460739332	10.85.153.129	10.85.135.96	ICMP	98	Echo (ping)

```
▸ Frame 1: 144 bytes on wire (1152 bits), 144 bytes captured (1152 bits) on interface wlp3s0, i
▸ Ethernet II, Src: 2c:9c:58:24:1c:21 (2c:9c:58:24:1c:21), Dst: e2:8d:90:ce:4b:61 (e2:8d:90:ce:
▸ Internet Protocol Version 4, Src: 10.85.153.129, Dst: 104.18.39.21
▸ Transmission Control Protocol, Src Port: 58608, Dst Port: 443, Seq: 1, Ack: 1, Len: 90
▸ Transport Layer Security
```

arp -a setelah ping

```
arz@RaidenArya:~$ arp -a
? (10.85.153.96) at d8:b3:2f:27:c2:bd [ether] on wlp3s0
_gateway (10.85.153.85) at e2:8d:90:ce:4b:61 [ether] on wlp3s0
RaidenArya (10.85.153.130) at <incomplete> on wlp3s0
```

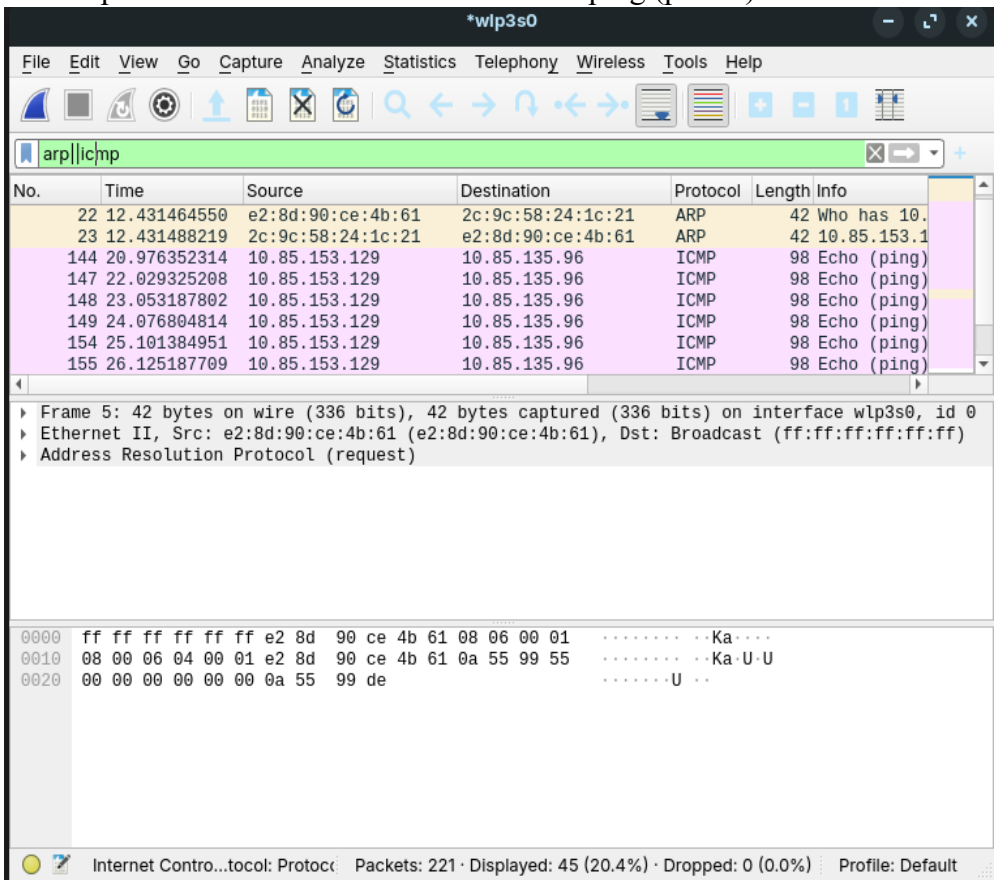
```
root@vian-ubuntu:~# arp -n
Address                  HWtype  HWaddress           Flags Mask            Iface
10.85.153.129            ether   2c:9c:58:24:1c:21   C                     wlp3s0
10.85.153.85             ether   e2:8d:90:ce:4b:61   C                     wlp3s0
10.85.153.130            ether   2c:9c:58:24:1c:21   C                     wlp3s0
```

Analisa:

- Saat sumber “tahu IP target tapi belum tahu MAC”, host akan mengirim ARP request broadcast (MAC tujuan FF:FF:FF:FF…) dan semua host LAN mengecek request itu; target membalas ARP reply.

- Setelah reply diterima, mapping IP→MAC disimpan di ARP cache sehingga arp -a berubah (muncul entri target).

C. Catat proses di wireshark setelah dilakukan ping (poin 5)



Analisa

- Perintah sama tapi hasil capture beda karena state ARP cache berbeda: sebelumnya cache kosong, setelah ping pertama cache sudah berisi mapping IP→MAC. (Makanya modul menanyakan "kenapa beda walau command sama").

D. Catat tabel arp cache yang tersimpan di /proc/net/arp

```
arz@RaidenArya:~$ cat /proc/net/arp
IP address       HW type     Flags       HW address            Mask     Device
10.85.153.96     0x1         0x2         d8:b3:2f:27:c2:bd     *        wlp3s0
10.85.153.85     0x1         0x2         e2:8d:90:ce:4b:61     *        wlp3s0
10.85.153.130    0x1         0x0         00:00:00:00:00:00     *        wlp3s0
```

Analisa:

- Ini versi file dari ARP cache yang sama, jadi isinya konsisten dengan arp -a, hanya format kolomnya lebih mentah.

E. Percobaan dengan pengurangan atau disable ARP cache (poin 7-8)

```
arz@RaidenArya:~$ sudo arp -d 10.85.153.96
No ARP entry for 10.85.153.96
arz@RaidenArya:~$ cat /proc/net/arp
IP address       HW type     Flags       HW address            Mask     Device
10.85.153.85     0x1         0x2         e2:8d:90:ce:4b:61     *        wlp3s0
10.85.153.130    0x1         0x0         00:00:00:00:00:00     *        wlp3s0
```

Jalankan arp -d untuk menghapus IP

```
arz@RaidenArya:~$ arp -a
arz@RaidenArya:~$
```

Tidak muncul ip

Analisa:

- entry target hilang, jadi ping berikutnya (kalau ARP masih aktif) akan memicu ARP request lagi.
- saat NOARP, host tidak bisa melakukan resolusi IP, padahal untuk mengirim frame Ethernet butuh MAC tujuan. Akibatnya komunikasi ke host tetangga (satu LAN) sering gagal.

F. Catat hasilnya dari poin 8

```
arz@RaidenArya:~$ sudo ifconfig wlp3s0 arp up
arz@RaidenArya:~$ sudo ifdown wlp3s0
arz@RaidenArya:~$ sudo ifup wlp3s0
arz@RaidenArya:~$ sudo ifconfig wlp3s0 arp up
arz@RaidenArya:~$ arp -a
_gateway (10.85.153.85) at e2:8d:90:ce:4b:61 [ether] on wlp3s0
? (10.85.153.96) at d8:b3:2f:27:c2:bd [ether] on wlp3s0
```

gunakan dhclient untuk mendapatkan IP dari server

```
arz@RaidenArya:~$ arp -a
_gateway (10.85.153.85) at e2:8d:90:ce:4b:61 [ether] on wlp3s0
RaidenArya (10.85.153.130) at <incomplete> on wlp3s0
? (10.85.153.96) at d8:b3:2f:27:c2:bd [ether] on wlp3s0
```

Analisa:

- interface dikembalikan normal sehingga ARP aktif lagi, host bisa belajar MAC target lagi lewat ARP.

G. Percobaan dengan penambahan ARP cache (poin 9)

```
root@vian-ubuntu:~# ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST>  mtu 1500
        inet 172.17.0.1  netmask 255.255.0.0  broadcast 172.17.255.255
        ether a6:a5:25:2f:d6:92  txqueuelen 0  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 36 overruns 0  carrier 0  collisions 0

eno1: flags=4099<UP,BROADCAST,MULTICAST>  mtu 1500
        ether bc:fc:e7:56:12:25  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 632  bytes 62326 (62.3 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 632  bytes 62326 (62.3 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

wlp3s0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.85.153.96  netmask 255.255.255.0  broadcast 10.85.153.255
        inet6 fe80::5ed5:eace:e1a5:21b8  prefixlen 64  scopeid 0x20<link>
        ether d8:b3:2f:27:c2:bd  txqueuelen 1000  (Ethernet)
        RX packets 6626  bytes 2919869 (2.9 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 4568  bytes 2546900 (2.5 MB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

```
arz@RaidenArya:~$ sudo arp -s 10.85.253.96 d8:b3:2f:27:c2:bd
```

```
arz@RaidenArya:~$ arp -a
_gateway (10.85.153.85) at e2:8d:90:ce:4b:61 [ether] on wlp3s0
? (10.85.253.96) at d8:b3:2f:27:c2:bd [ether] PERM on eno1
RaidenArya (10.85.153.130) at <incomplete> on wlp3s0
? (10.85.153.96) at d8:b3:2f:27:c2:bd [ether] on wlp3s0
arz@RaidenArya:~$ ping 10.85.153.96
PING 10.85.153.96 (10.85.153.96) 56(84) bytes of data.
64 bytes from 10.85.153.96: icmp_seq=1 ttl=64 time=137 ms
64 bytes from 10.85.153.96: icmp_seq=2 ttl=64 time=159 ms
64 bytes from 10.85.153.96: icmp_seq=3 ttl=64 time=184 ms
64 bytes from 10.85.153.96: icmp_seq=4 ttl=64 time=208 ms
64 bytes from 10.85.153.96: icmp_seq=5 ttl=64 time=127 ms
^C
--- 10.85.153.96 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4004ms
rtt min/avg/max/mdev = 127.337/163.012/207.953/29.700 ms
```

Analisa:

- Ini memaksa mapping IP tanpa perlu ARP request.
- Kalau MAC yang dimasukkan salah, paket bisa salah tujuan.

H. Catat hasil pada poin 10

```
11:16:28.729731 IP 155.102.28.26.443 > 10.85.153.129.38708: Flags [P.], seq 445085483:445085507, a
ck 793699973, win 155, length 24
11:16:28.729781 IP 10.85.153.129.38708 > 155.102.28.26.443: Flags [.], ack 24, win 547, length 0
11:16:28.729792 IP 155.102.28.26.443 > 10.85.153.129.38708: Flags [F.], seq 24, ack 1, win 155, le
ngth 0
11:16:28.771098 IP 10.85.153.129.38708 > 155.102.28.26.443: Flags [.], ack 25, win 547, length 0
11:16:28.928262 IP 155.102.28.22.443 > 10.85.153.129.37398: Flags [P.], seq 3350805070:3350805094,
 ack 2844682501, win 203, length 24
11:16:28.928327 IP 10.85.153.129.37398 > 155.102.28.22.443: Flags [.], ack 24, win 573, length 0
11:16:28.928348 IP 155.102.28.22.443 > 10.85.153.129.37398: Flags [F.], seq 24, ack 1, win 203, le
ngth 0
11:16:28.969103 IP 10.85.153.129.37398 > 155.102.28.22.443: Flags [.], ack 25, win 573, length 0
```

```
arz@RaidenArya:~$ sudo tcpdump -n -t -e -i wlp3s0
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on wlp3s0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
fe:75:c5:73:a5:43 > 01:00:5e:00:00:fb, ethertype IPv4 (0x0800), length 122: 10.85.153.147.5353 > 2
24.0.0.251.5353: 0 [3q] PTR (QM)? _rdlink._tcp.local. PTR (QM)? _companion-link._tcp.local. PTR (Q
M)? lb._dns-sd._udp.local. (80)
2c:9c:58:24:1c:21 > e2:8d:90:ce:4b:61, ethertype IPv4 (0x0800), length 136: 10.85.153.129.37784 >
31.13.95.60.5222: Flags [P.], seq 324132764:324132834, ack 2145632903, win 656, options [nop,nop,T
S val 334916325 ecr 413766947], length 70
e2:8d:90:ce:4b:61 > 2c:9c:58:24:1c:21, ethertype IPv4 (0x0800), length 66: 31.13.95.60.5222 > 10.8
5.153.129.37784: Flags [.], ack 70, win 235, options [nop,nop,TS val 413776068 ecr 334916325], len
gth 0
```

Analisa:

- -n = jangan resolve DNS/hostname (lebih cepat & jelas IP mentah)
- -t = hilangkan timestamp
- -e = tampilkan header Ethernet (MAC sumber/tujuan) → penting untuk analisa Layer 2
- -i eth0 = pilih interface
- -X = tampilkan payload hex+ASCII (untuk lihat isi data paket)

I. Catat hasil pada poin 11 (tcpdump)

```
arz@RaidenArya:~$ sudo tcpdump -n -e -i wlp3s0
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on wlp3s0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
11:21:28.741384 2c:9c:58:24:1c:21 > e2:8d:90:ce:4b:61, ethertype IPv4 (0x0800), length 131: 10.85.
153.129.37784 > 31.13.95.60.5222: Flags [P.], seq 324135554:324135619, ack 2145637012, win 656, op
tions [nop,nop,TS val 335076483 ecr 413923244], length 65
11:21:28.746401 2c:9c:58:24:1c:21 > e2:8d:90:ce:4b:61, ethertype IPv4 (0x0800), length 150: 10.85.
153.129.37784 > 31.13.95.60.5222: Flags [P.], seq 65:149, ack 1, win 656, options [nop,nop,TS val
335076488 ecr 413923244], length 84
11:21:29.045470 e2:8d:90:ce:4b:61 > 2c:9c:58:24:1c:21, ethertype IPv4 (0x0800), length 66: 31.13.9
5.60.5222 > 10.85.153.129.37784: Flags [.], ack 65, win 245, options [nop,nop,TS val 413936527 ecr
 335076483], length 0
11:21:29.045521 e2:8d:90:ce:4b:61 > 2c:9c:58:24:1c:21, ethertype IPv4 (0x0800), length 66: 31.13.9
5.60.5222 > 10.85.153.129.37784: Flags [.], ack 149, win 245, options [nop,nop,TS val 413936533 ec
r 335076488], length 0
11:21:33.127725 fe:75:c5:73:a5:43 > 01:00:5e:00:00:fb, ethertype IPv4 (0x0800), length 122: 10.85.
153.147.5353 > 224.0.0.251.5353: 0 [3q] PTR (QM)? _rdlink._tcp.local. PTR (QM)? _companion-link._t
cp.local. PTR (QM)? lb._dns-sd._udp.local. (80)
11:21:33.130692 fe:75:c5:73:a5:43 > 33:33:00:00:00:fb, ethertype IPv6 (0x86dd), length 142: fe80::
1c23:8490:a6cb:738a.5353 > ff02::fb.5353: 0 [3q] PTR (QM)? _rdlink._tcp.local. PTR (QM)? _companio
n-link._tcp.local. PTR (QM)? lb._dns-sd._udp.local. (80)
```

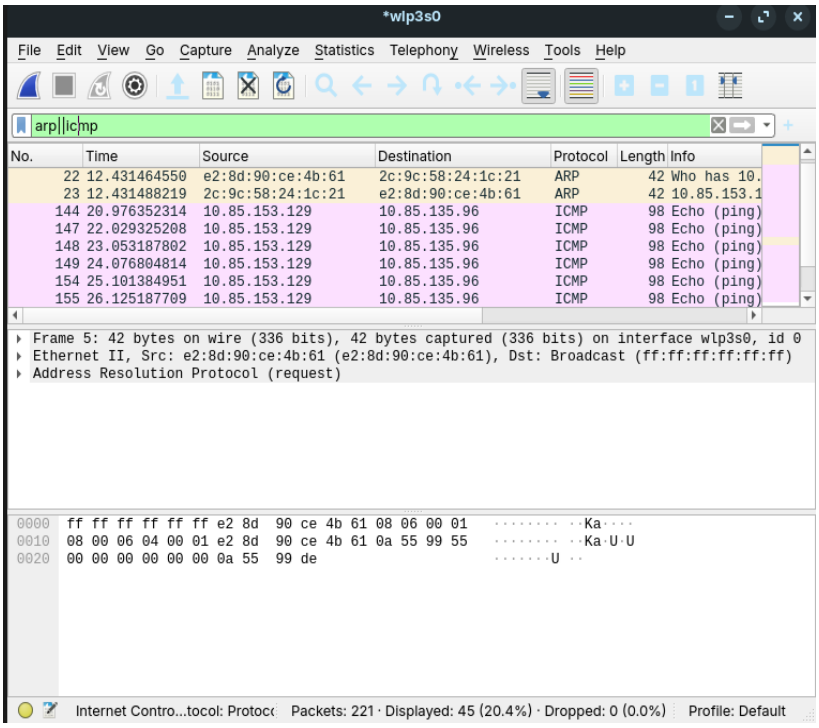
```
arz@RaidenArya:~$ ping 10.85.153.96 -c 2
PING 10.85.153.96 (10.85.153.96) 56(84) bytes of data.
64 bytes from 10.85.153.96: icmp_seq=1 ttl=64 time=5.04 ms
64 bytes from 10.85.153.96: icmp_seq=2 ttl=64 time=17.7 ms

--- 10.85.153.96 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 5.039/11.351/17.663/6.312 ms
```

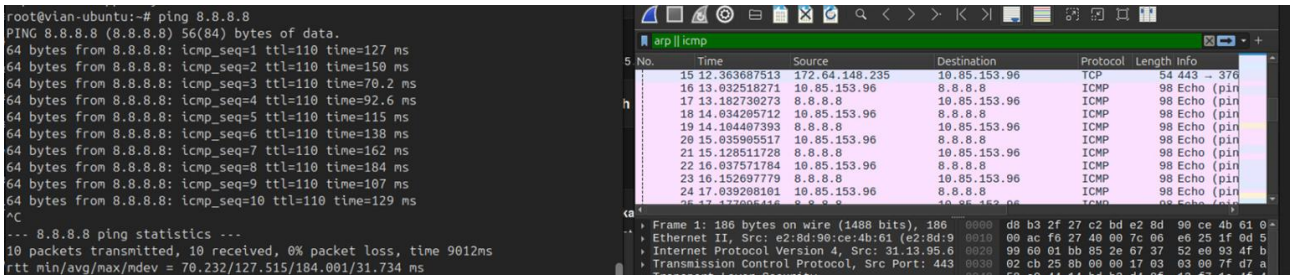
```
arz@RaidenArya: ~
11:24:47.079483 IP 114.125.99.225.https > RaidenArya.51820: UDP, length 1232
11:24:47.079502 IP 114.125.99.225.https > RaidenArya.51820: UDP, length 1232
11:24:47.079509 IP 114.125.99.225.https > RaidenArya.51820: UDP, length 1232
11:24:47.080053 IP 114.125.99.225.https > RaidenArya.51820: UDP, length 1232
11:24:47.080077 IP 114.125.99.225.https > RaidenArya.51820: UDP, length 1232
11:24:47.080085 IP 114.125.99.225.https > RaidenArya.51820: UDP, length 1232
11:24:47.080183 IP RaidenArya.51820 > 114.125.99.225.https: UDP, length 35
11:24:47.080352 IP 114.125.99.225.https > RaidenArya.51820: UDP, length 1232
11:24:47.080390 IP 114.125.99.225.https > RaidenArya.51820: UDP, length 1232
11:24:47.080398 IP 114.125.99.225.https > RaidenArya.51820: UDP, length 1232
11:24:47.085405 IP 114.125.99.225.https > RaidenArya.51820: UDP, length 1232
11:24:47.085424 IP 114.125.99.225.https > RaidenArya.51820: UDP, length 1232
11:24:47.085595 IP 114.125.99.225.https > RaidenArya.51820: UDP, length 1232
11:24:47.085895 IP 114.125.99.225.https > RaidenArya.51820: UDP, length 1200
11:24:47.086914 IP RaidenArya.51820 > 114.125.99.225.https: UDP, length 36
11:24:47.102309 IP 114.125.99.225.https > RaidenArya.51820: UDP, length 48
^C11:24:47.401115 IP 10.85.153.147 > all-routers.mcast.net: igmp leave mdns.mcast.net

571 packets captured
580 packets received by filter
0 packets dropped by kernel
```

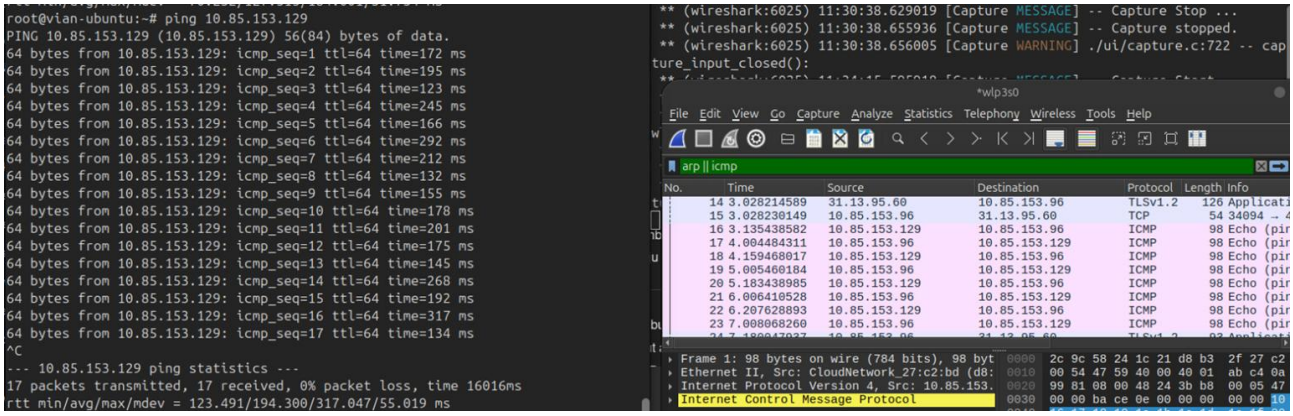
Analisa:

- sekarang tabel ARP memuat entri IP target menjadi MAC target.
- PC tidak mencari MAC milik 8.8.8.8.
- PC akan mengirim frame ke MAC default gateway (next hop), jadi yang biasanya muncul di ARP adalah gateway.
- browser memicu banyak traffic (DNS lookup, TCP handshake, lalu TLS/HTTPS), sehingga output tcpdump jadi ramai.

J. Catat hasil pada poin 12 (wireshark)







Analisa:

- Ping 1 subnet (ke 10.85.153.130):
  dst IP = 10.85.153.130
  dst MAC = MAC target (aa:bb:cc:dd:ee:ff)
- Ping luar subnet (mis. 8.8.8.8):
  dst IP = 8.8.8.8
  dst MAC = MAC gateway, bukan MAC 8.8.8.8
- Ini yang membuktikan: alamat L3 (IP) end-to-end menuju tujuan, sedangkan alamat L2 (MAC) hanya menuju next hop.